

SESSION 4 · HANDS-ON LAB

COMPLETE GUIDED TUTORIAL

Verifying Synthetic and Manipulated Media

A browser-only laboratory in reverse image search, InVID keyframes, metadata reading, and the professional fact-checking workflow

MISSION: Deliver five calibrated verdicts on a mixed exhibit pack — and defend the one that is genuine.

Duration	Coding level	Installation
Two activities · 25 minutes each	No coding at all	Browser + one free plugin (InVID-WeVerify)

Participant: _____ **Date:** _____

Companion resources supplied with this manual

1. Exhibit pack folder (link on your handout) — five media items V1–V5 plus one original photo file with metadata intact.
2. Shared scoreboard Sheet (edit link on your handout) — one row per pair; the debrief runs from this board.
3. InVID-WeVerify browser plugin (install link on your handout) — free, built with professional fact-checkers.

Safety and integrity rule

The exhibit pack mixes miscaptioned, fabricated, AI-generated AND genuine items, assembled for training. Do not forward, post, or share any exhibit outside this room — a training fake released into a family WhatsApp group becomes real misinformation. Do not upload personal or identifying images to any tool during this lab. Verifying media in a live case requires an authorised process and evidence preservation before analysis.

1 • ACTIVITY OVERVIEW

What You Will Do

You staff a district verification desk. Five items have arrived from monitoring and public complaints — a flood photo, a circulating “official” memo, a protest image, a 40-second video clip, and one picture that looks too dramatic to be true. Some are lies about real pixels; some are honest pixels under lying captions; at least one is entirely genuine. You will trace each item’s history with reverse image search and keyframes, inspect what pixels and metadata can actually show, run the full professional fact-checking workflow, and file a calibrated verdict for every item — including the courage to call the true one true.

Questions your investigation must answer

- ☐ Where and when did each image or clip FIRST appear — and what did its original caption say?
- ☐ What does “no search results” actually mean — and what must you never conclude from it?
- ☐ Which defects in a fabricated document survive a glance but not a magnifier?
- ☐ Has any exhibit already been checked by India’s fact-checking ecosystem — and how do you find out in two minutes?
- ☐ Can you verify a genuine item as genuine — with evidence, not vibes — when its content is shocking?

Process you will follow

1. Read	2. Vote	3. Trace	4. Inspect	5. Extract	6. Check	7. Verdict	8. Report
Claim cards	Cold open	Reverse search	Keyframes + magnifier	The checkable claim	Fact-check sites	Calibrated language	Board + debrief

Timing and deliverables

Time	Stage	Output
Before Activity 1	Read Sections 1–5; cold-open vote on V1	Committed written vote + reason
Activity 1 • 0–12 min	Part A: reverse image search on V1, V3, V5	Earliest-appearance records + draft verdicts
Activity 1 • 12–25 min	Part B: InVID keyframes (V4), magnifier (V2), metadata demo	Defect lists + context checks
Debrief A • 15 min	The reveal, from the scoreboard	Corrected verdicts + the V3 / V5 lessons
Activity 2 • 0–15 min	Part A: full fact-checking workflow, all five	Verdict + confidence + evidence + falsifier, per item
Activity 2 • 15–25 min	Part B: readiness audit + the dividend question	Organisation score /3 + one hard answer
Debrief B • 15 min	Trust round, from the scoreboard	Personal protocol, stated in one breath

2 · EXHIBIT PACK — FIVE CLAIMS UNDER TEST

Read the Claims Before Touching Any Tool

The media files live in the exhibit folder on your handout link. This section reproduces each item's claim card — the caption exactly as it circulates. Remember the discipline: the claim under test is the pair (media + caption), not the pixels alone. A true photo under a false caption is a false claim.



EXHIBIT V1 | Image + claim — “Flood devastation, this week”

[image file: V1_flood.jpg – in the exhibit folder]

Circulating caption:

“BREAKING: Highway near Rajgarh completely underwater since last night. Rescue teams overwhelmed. Media is silent – share this before it is taken down!”

What you receive — One dramatic flood photograph plus the caption above, as forwarded.

EXHIBIT V2 | Screenshot — “Leaked official circular”

[image file: V2_memo.png – in the exhibit folder]

Text visible in the screenshot:

“URGENT CIRCULAR – Office of the District Collector, Meridianpur. In view of severe water contamination detected in the municipal supply, all schools, colleges and Anganwadi centres shall remain closed from Monday until further orders. Citizens are advised to withdraw sufficient cash as banking operations may be restricted. Ref No: DC/MPR/2024/0113.

By order.”

Forwarded with: “Forwarded as received. Please inform family.”

What you receive — A screenshot of the circular as it circulates on messaging apps. Meridianpur is the fictional district of this course universe.

EXHIBIT V3 | Image + claim — “Protest crowd”

[image file: V3_protest.jpg – in the exhibit folder]

Circulating caption:

“Massive crowd on the flyover tonight against the new toll.
They will not show this on TV.”

What you receive — One striking night photograph of a large crowd, plus the caption.

EXHIBIT V4 | Video — “Official’s statement” (40 seconds)

[video file: V4_statement.mp4 – in the exhibit folder]

Circulating caption:

“Water department officer admitting contamination in OUR
district yesterday. They knew. Listen from 0:22.”

What you receive — A 40-second clip of a real press briefing, plus the caption. Your job is the pair — does the caption’s where and when survive the frames?

EXHIBIT V5 | Image + claim — “Shocking, but real?”

[image file: V5_sky.jpg – in the exhibit folder]

Circulating caption:

“This is a real, unedited photo of the sky before a storm.
No filter. Nature is terrifying.”

What you receive — One image whose content most people refuse to believe on sight. The claim asserts authenticity — test exactly that.

Human-first triage (before any tool)

Gut verdicts now, in ink. The debrief compares the room’s instincts — especially on V5 — against what the evidence later showed.

Exhibit	Gut verdict: True / False / Can’t tell	Confidence L / M / H	The one thing driving your vote
V1			
V2			
V3			
V4			
V5			

3 · YOUR VERIFICATION BENCH

Tools and the Question Each One Answers

Tool	Role in this lab	The question it helps answer
Google Lens / images.google.com	Primary reverse image search	Where has this image appeared before — and when was the EARLIEST time?
TinEye (tineye.com)	Cross-check with oldest-first sorting	Does a second index confirm the earliest appearance?
InVID-WeVerify plugin	Keyframes, magnifier, metadata, contextual queries	What do the frames, the fine detail, and the file itself say?
Fact-checking sites (Section 4 list)	India's working correction infrastructure	Has this claim — or this exact image — already been checked?
Shared scoreboard Sheet	The room's evidence board	What did the whole room conclude — and where does it disagree?
Your judgement	The only accountable instrument	What verdict will a named human sign, in defensible language?

Concepts in plain language

SIFT — the mindset before any pixel-gazing

Stop (pause the emotional reflex the content was built to trigger) → Investigate the source (who published this; what is their record and incentive) → Find better coverage (is anyone credible reporting it — absence is a finding) → Trace to the original (the unedited item, in its first context, with its first date).

Cheapfake vs deepfake vs full synthesis

Cheapfake: real media, false context — an old photo under a new caption; no AI required, and historically the majority of viral visual misinformation. Manipulated: a real base, surgically altered. Fully synthetic: generated from nothing, so it has no history to find. Different fakes fall to different tools — misclassify the fake and you pick the wrong forensics.

Provenance beats pixels

No pixel analysis detects a true photo under a false caption — only its history does. Ask where it came from before asking what is wrong with it. This is why reverse image search, not a deepfake detector, is the workhorse of this lab.

Zero matches ≠ authentic

A fully generated image has no prior history — but so does a photo that is too new, too cropped, or simply not indexed. “No results” supports at most “no indexed history found”; the correct verdict language is “consistent with AI generation”, never “proven fake”. This is the V3 trap, named in advance.

Calibrated verdict language — the quality bar

“Miscaptioned — original is a 2019 flood photograph (evidence: earliest indexed appearance, original caption)” beats “fake”. And the crucial asymmetry: “no evidence of manipulation found” is NOT “authentic” — absence of findings is a bounded statement about your search, never a certificate.

The liar's dividend

As deepfake awareness grows, “that is a deepfake” becomes a universal defence against genuine evidence. Verification is therefore no longer only about catching fakes — it is about defending the real. Exhibit V5 exists because of this concept.

4 • BENCH SETUP

Browser, Plugin, Bookmarks: From Zero to a Working Desk

Step 1. Open the plugin link on your handout in Chrome, Edge, or Firefox and add the extension. Pin it to the toolbar. One working install per pair is enough.

Step 2. Verify reverse image search. Right-click any image on any webpage. If “Search image with Google” (or “Search with Google Lens”) appears, you are ready. Fallback that always works: open images.google.com and use the camera icon to upload or paste an image.

Step 3. Bookmark the fact-checkers. Open and bookmark each site in the table below — you will race through them in Activity 2.

Step 4. Open the exhibit folder and claim your scoreboard row. Confirm all five media files open, type both partners’ names in the next free row, and leave both tabs open.

The fact-checking shelf (bookmark all)

Organisation	Address	Strength in this lab
Alt News	altnews.in	Indian viral claims, image and video debunks, Hindi + English
BOOM	boomlive.in	Indian misinformation, strong on WhatsApp-circulated media
Factly	factly.in	Indian claims with document and data verification
PIB Fact Check	pib.gov.in (Fact Check section)	Claims about Government of India circulars and schemes — exactly the V2 category
AFP Fact Check	factcheck.afp.com	International wire-service checks, searchable by image theme
Snopes	snopes.com	Long-running international archive — useful for the V5 category of “unbelievable but real”

What the InVID-WeVerify screen means

Tool tab	What it does	Use in this lab
Keyframes	Splits a video into representative still frames	Turn V4 into images, then reverse-search each frame
Magnifier	Zoom and enhancement for fine detail	Inspect V2’s fonts, seal, alignment, reference number
Metadata	Reads EXIF from an image or video FILE	Run on the supplied original photo; compare with a platform-stripped copy
Image search	One-click reverse search across engines	Faster route to Lens, Yandex, Bing per frame

If the plugin will not install (managed lab machines)

Everything has a manual fallback: pause the video at a distinctive moment, take a screenshot, and reverse-search the screenshot — that is keyframing by hand. Zoom the memo with your browser (Ctrl and +) instead of the magnifier. Read metadata with any free web EXIF viewer, on the ORIGINAL file only — and never upload personal images.

Successful setup checklist

- ☐ InVID-WeVerify pinned (or fallback plan noted).
- ☐ Right-click reverse image search works, or images.google.com is open.
- ☐ All six fact-checking sites bookmarked.
- ☐ All five exhibit files open from the folder.
- ☐ Our scoreboard row has our names in it.

5 · TOOL TUTORIAL

Reverse Image Search: The False-Context Killer

The single most valuable workflow in media verification, and the one you will run first. Its power is history: a recycled photo dies the moment its earlier life surfaces.

Step	Exact action	What you are looking for
1. Search	Right-click the exhibit → “Search with Google Lens”, or upload at images.google.com	Exact matches and visually similar results
2. Widen	Switch result tabs; scroll past shopping junk; open “Find image source” where offered	Pages that embed this exact image
3. Date-hunt	Scan result dates; open promising older pages; note publication dates	The EARLIEST indexed appearance — the whole game
4. Cross-check	Upload the same image to tineye.com → sort by “Oldest”	A second index confirming (or beating) your earliest date
5. Read the original	Open the earliest source properly	Original caption, original date, original location
6. Compare	Set the original context beside the circulating caption	Does the claim survive the image’s actual history?

Craft notes from working fact-checkers

Crop and search regions separately when the full image stalls — a face, a signboard, a skyline. Try the mirrored version; recyclers flip images to dodge matching. Screenshots of screenshots still match — persistence beats sophistication. And sort TinEye by oldest, not best match: you are hunting a date, not a lookalike.

Metadata in two honest sentences

When present, EXIF is a provenance goldmine: camera model, timestamp, GPS, editing-software traces. But every major platform strips EXIF on upload — so absence is normal, not suspicious, and metadata survives only on original files obtained from a device, an email attachment, or a direct transfer.

6 · ACTIVITY 1 — TRACE THE MEDIA (25 MIN)

Part A · Reverse Image Verification — V1, V3, V5 (12 min)

Run the Section 5 workflow on each of the three images. Budget roughly four minutes each; V1 should fall fastest. Record everything in the tables — an undocumented search never happened.

Record — per exhibit	V1	V3	V5
Any matches found? (Y / N)			
Earliest appearance — date			
Earliest appearance — source / site			
Original caption or context (short)			
Does the circulating claim survive? (Y / N / Cannot say)			
The V3 discipline — interpreting absence If V3 returns nothing, write down what “no history” can mean — too new, too cropped, not indexed, or never real — and which additional check each possibility demands. Then draft the verdict in bounded language. The debrief grades your sentence, not just your answer.			

V3 draft verdict, in calibrated language:

V5 draft verdict, in calibrated language (“shocking” and “false” are different properties):

Part B · InVID and Metadata — V2, V4 (13 min)

V4 — keyframe the video (7 min). InVID → Keyframes → load V4_statement.mp4 → reverse-search the three most distinctive frames. Then run the context checks below.

V4 context check	What the frames show	Does the caption survive?
Earliest appearance of any frame (date, source)		
Venue / backdrop — matches “our district yesterday”?		
Language, banners, insignia visible in frame		
Weather / season vs the claimed date		

V2 — magnify the memo (6 min). InVID Magnifier (or browser zoom). Work the checklist; every box you tick is one line of your verdict’s evidence.

- ☐ Fonts: does any line differ in typeface, weight, or size from the rest?
- ☐ Alignment: do margins, headers, and the seal sit where an official template would put them?
- ☐ Seal / emblem: crisp official artwork, or a soft generic clip-art paste?
- ☐ Reference number: is DC/MPR/2024/0113 internally consistent with the memo’s own claimed date and series?
- ☐ Register: does the wording read like administrative drafting — or like panic bait (“withdraw sufficient cash”)?
- ☐ PIB / district precedent: have real circulars of this type ever issued this instruction, in this format?

Metadata demo (with remaining time). Run InVID Metadata (or a web EXIF viewer) on the supplied ORIGINAL photo file from the exhibit folder, then on any exhibit downloaded from the messaging-app copy. Note what survives in one and vanishes in the other — and remember which direction is normal.

Scoreboard entry — end of Activity 1

Enter for each of V1–V5: verdict + confidence, and the single strongest piece of evidence, tagged by type — PROV (provenance / earliest appearance), TECH (pixels, fonts, metadata), or CTX (context checks). The debrief counts which evidence types actually carried the room’s correct verdicts.

7 · ACTIVITY 2 — THE FULL FACT-CHECKING WORKFLOW (25 MIN)

Part A · All Five, End to End (15 min)

Step 1. Extract the claim. For each exhibit, write ONE checkable sentence — strip the outrage, keep the assertion. “This photo shows flooding near Rajgarh this week”, not “media is silent!”

Step 2. Search the checkers. Race the bookmarked shelf. Query with two or three distinctive keywords, not the full caption; try both English and Hindi terms; for images, also try each site’s own search with the image’s subject (“green sky storm”, “flood highway photo old”).

Step 3. Read laterally. Leave the exhibit; ask what credible, independent coverage says about the underlying event. On a claim this dramatic, absence of any corroboration is itself a finding — write it down as one.

Step 4. File the verdict. Complete the worksheet — including the falsifier column: what specific evidence would change your mind. A verdict without a falsifier is a belief, not a finding.

Exhibit	Claim in one sentence	Verdict + confidence	Strongest evidence	What would change my mind
V1				
V2				
V3				
V4				
V5				

The V5 test — defend the real

One exhibit is genuine. Verifying it as genuine — earliest appearances traced, consistent independent coverage found, phenomenon explained — is the liar’s-dividend skill this whole session builds toward. If your worksheet proves the true one true, you are doing the full job, not half of it.

Part B · Readiness Audit and the Hard Question (10 min)

Score your organisation — agency, university, company, or unit — on the three defences that beat the most famous deepfake attack on record. One point each; most organisations score one of three.

The trio	Question	Score 0 / 1
Prebunked audience	Have your people been told, in advance, what fabricated content targeting you would look like?	
Authenticated rapid channel	If you had to rebut a fake in two hours, is there an established channel your audience already trusts and checks?	
Response playbook	Does a written plan exist: evidence capture, platform takedown, counter-communication, in that order?	

Total: ____ / 3. Which is your missing pair, and who owns fixing it?

The dividend question — a genuine, incriminating recording surfaces in one of your cases, and the subject declares it a deepfake. What must your side now prove, and could you, today?

8 • FINAL REPORT AND DEBRIEF

Close the Row, Then Close the Loop

Scoreboard row — completion check

- ☐ V1–V5 verdicts with confidence entered, each with an evidence tag (PROV / TECH / CTX).
- ☐ Falsifier written for every verdict.
- ☐ Fact-check hit or miss recorded per exhibit.
- ☐ Readiness score /3 and missing-pair answer entered.
- ☐ Dividend answer typed in the last column.

Your personal protocol, in one breath

Complete the sentence you will actually use, starting tonight, when shocking media reaches your phone:

“Before I forward anything, I will — (1) ... (2) ... (3) ...”

Recommended action options (organisational)

- ☐ Adopt SIFT + reverse search as the desk standard for all incoming media claims.
- ☐ Add “video call” and “voice note” to the channels that can NO LONGER verify identity for payments.
- ☐ Stand up the missing element(s) of the trio: prebunking, authenticated channel, response playbook.
- ☐ Require calibrated verdict language in all reports — ban bare “fake” / “real”.
- ☐ Route suspected government-circular fakes to PIB Fact Check and the district PRO the same day.

One-minute reflection

Which exhibit fooled you longest — and which evidence type finally settled it?

What is one claim you personally forwarded in the past that would not have survived today’s workflow?

9 · TROUBLESHOOTING, GLOSSARY AND SIGN-OFF

When Something Goes Wrong

Problem	Likely cause	Exact action
Reverse search returns shopping junk	Similarity match dominating	Crop to the distinctive region and search again; try the mirrored image; switch to TinEye
No results at all	Not indexed — or never real	Record “no indexed history”; NEVER conclude authentic or fake from absence alone; run the extra checks from the V3 box
InVID will not install	Managed machine / blocked store	Manual fallback: pause → screenshot → reverse-search frames; browser zoom for the memo
Keyframes tab rejects the file	Local file quirk	Play the clip, pause at three distinct moments, screenshot each, search those
EXIF viewer shows nothing	Platform stripped it on upload	Normal, not suspicious — metadata lives only on originals; use the supplied original file
Fact-check sites slow or down	Free-tier load	Google the keywords + site name (e.g. flood photo old altnews); the cached result answers you
Two pairs claim one scoreboard row	Race condition	Take the next free number; never overwrite
Online participant cannot speak	Hybrid audio limits	Type in meeting chat; the instructor reads chat rows aloud

Quick glossary

Term	Meaning in this lab
Cheapfake	Real media in false context — no AI required, and the most common viral fake
Deepfake	AI-manipulated media of a real person — face swap, lip-sync, cloned voice
Full synthesis	Media generated from nothing; it has no history for search to find
Provenance	The item’s documented history: first appearance, first context, first date
SIFT	Stop · Investigate the source · Find better coverage · Trace to the original
Keyframe	A representative still extracted from video so image tools can work on it
EXIF	Metadata inside an original media file: device, timestamp, GPS, edit traces
Earliest appearance	The oldest indexed sighting — the decisive fact against recycled media
Lateral reading	Leaving the item to check what independent, credible sources say
Miscontextualised	Honest pixels under a lying caption — the workhorse of visual misinformation
Liar’s dividend	Genuine evidence dismissed as “probably a deepfake” — the deeper injury
Prebunking	Inoculation before exposure: naming the technique before the fake arrives
Content credentials (C2PA)	Cryptographically signed capture-and-edit history travelling with a file
Calibrated verdict	A conclusion with evidence, confidence, and stated limits — never bare “fake”

Completion checklist

- ☐ I voted on gut first, and kept the vote for the debrief.
- ☐ I hunted earliest appearances instead of admiring pixels.
- ☐ I wrote what “no results” can and cannot mean before concluding anything.

- ☐ I filed every verdict with evidence, confidence, a stated limit, and a falsifier.
- ☐ I verified the genuine item as genuine — with evidence, not vibes.
- ☐ My scoreboard row is complete — the debrief runs on it.